You'll see how to write basic and advanced Active Record queries, as well as write SQL within Active Record. Active Record doesn't limit you to writing SQL queries solely in Ruby, so you've got options.

In Active Record, fragments of SQL queries like filter conditions in a WHERE clause are called *Named Scopes*. The Named Scopes pattern provides a conventional technique to identify most of the "filtering code" by reading through application models. Named Scopes can be shared among models as well and there can be default scopes that get applied to all queries for a particular model.

One example where common filtering occurs and reuse helps is with "soft deletes." In a conventional soft deletes design, users delete records which are then hidden from the application but not physically removed as rows. Instead they have a deleted_at timestamp column that's set. This pattern is implemented in a common location as Ruby code leveraging the Active Record Named Scopes and Default Scope concepts.

Active Record allows a developer to start with a model class and gradually add methods to it. For this chainable design, query generation is deferred until the end, which is called lazy evaluation.

While this design is a good developer experience, it comes with a trade-off: developers can easily write inefficient queries, sometimes unknowingly.

Before you can fix inefficient queries, you'll need to see where they are coming from in the application.

Query Logs To Connect SQL to App Code

Although you may analyze slow queries in an Application Performance Monitoring (APM) tool, when analyzing SQL queries in the postgresql.log you can trace them back to the Active Record that generated them.

In this section, you'll learn how to link SQL queries to application code. Prior to Rails version 7, third party gems fulfilled this purpose, but eventually support was added directly to Rails for this very useful functionality.

Prior to Rails version 7, the Marginalia[1] gem added comments to SQL queries that showed where in the application the query was generated. The Marginalia comment lists the application name, controller name, and controller action name. An example comment looks like this:

```
/*application:Rideshare,controller:trips,action:index*/
```

1. https://github.com/basecamp/marginalia